

COMPUTING & INFORMATION SCIENCE DEPARTMENT
SCO400_SIT400_SCO444 – PROJECT

Concept Paper

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Proposed Project Title:

Design and Development of a Competency-Based Education (CBE) online System for Momokoro School

Problem Description and Justification:

Momokoro School is transitioning to Kenya's Competency-Based Education curriculum but lacks a centralized digital platform to support modern learning and administration. Students have difficulty accessing structured notes, quizzes, and progress-tracking tools, leading to fragmented learning and inconsistent content delivery. Teachers face challenges in sharing resources, organizing lesson plans, and monitoring learner progress. Administrators rely on manual processes to approve content and track school-wide learning activities, which reduces efficiency. A dedicated CBE online system is essential to unify these processes. It will enhance learner engagement, provide clear visibility of individual progress, and simplify resource management. A mobile-first design ensures accessibility for all learners and teachers, many of whom primarily use smartphones.

Proposed Solution:

Develop a **web-based CBE System** for Momokoro School featuring: **Learner Dashboard:** Centralized notes hub, subject-based quizzes with instant feedback, leaderboards, and personalized progress tracking. **Teacher Dashboard:** Tools to upload and manage resources, access schemes of work, and create or download lesson plans with an approval workflow. **Admin Panel:** User management, content approval, and activity oversight for administrators.

Proposed Tools for Solution:

Frontend: React.js with Tailwind CSS for responsive, mobile-first user interfaces.

Backend: Django (Python) for scalable RESTful APIs and role-based authentication.

Database: PostgreSQL for reliable storage of notes, quizzes, and user data.

Authentication: Firebase Authentication for secure user login and management.